

term	estimate	std.error	statistic	p.value
(Intercept)	2370.51090	182.29049	13.004030	0.0000012
X	-22.97465	11.99887	-1.914735	0.0918583

Properties of Least Squares Estimators

We can re-write the estimate of $\hat{\beta}_1$ as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

We note that $\hat{\beta}_0$ and β_1 are a linear combination of the observations of y_i

For instance

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^n c_i y_i$$

where $c_i = \frac{x_i - \bar{x}}{S_{xx}}$

Unbiasedness of the estimators

The least squares estimators $\hat{\beta}_0$ and β_1 are **unbiased estimators** of the model parameters β_0 and β_1 . To show this for $\hat{\beta}_1$ consider

$$\begin{aligned} E(\hat{\beta}_1) &= E\left(\sum_{i=1}^n c_i y_i = \sum_{i=1}^n c_i E(Y_i)\right) \\ &= \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^n c_i + \beta_1 \sum_{i=1}^n c_i x_i \end{aligned}$$

The terms $\sum_{i=1}^n c_i = 0$ and $\sum_{i=1}^n c_i x_i = 1$

Thus $E[\hat{\beta}_1] = \beta_1$

Similarly we may show that $\hat{\beta}_0$ is unbiased estimator of β_0

Consider

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\begin{aligned} E[\hat{\beta}_0] &= E[\bar{y} - \hat{\beta}_1 \bar{x}] \\ &= \bar{y} - \beta_1 \bar{x} \\ &= \beta_0 + \beta_1 \bar{x} - \beta_1 \bar{x} \\ &= \beta_0 \end{aligned}$$

Variances of the estimators

The variance of $\hat{\beta}_1$ is

$$Var(\hat{\beta}_1) = Var\left(\sum_{i=1}^n c_i y_i\right) = \sum_{i=1}^n c_i^2 Var(y_i) \quad (6)$$

because the observations y_i are uncorrelated and so the variance of the sum is just the sum of the variances.

The variances of each term in the sum is $c_i^2 Var(y_i)$ and we have assumed that $Var(y_i) = \sigma^2$ consequently

$$Var(\hat{\beta}_1) = \sigma^2 \sum_{i=1}^n c_i^2 = \frac{\sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}$$

The variance of $\hat{\beta}_0$ is

$$\begin{aligned} Var(\hat{\beta}_0) &= Var(\bar{Y} - \hat{\beta}_1 \bar{x}) \\ &= Var(\bar{y}) + \bar{x}^2 Var(\hat{\beta}_1) - 2\bar{x} Cov(\bar{y}, \hat{\beta}_1) \end{aligned}$$

Now the variance of $\bar{y}$ is $\frac{\sigma^2}{n}$ and the covariance between $\bar{y}$ and $\hat{\beta}_1$ can be shown to be zero
Thus

$$Var(\hat{\beta}_0) = Var(\bar{y}) + \bar{x}^2 Var(\hat{\beta}_1) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right]$$

Gauss-Markov Theorem

For the simple linear regression model with the assumptions that $E(\epsilon) = 0$, $Var(\epsilon) = \sigma^2$ and uncorrelated errors the least squares estimators are unbiased and have minimum variance when compared with all other unbiased estimators that are linear combination of y_i .

We say that the least squares estimators are **best linear unbiased estimators** where best implies minimum variance.

Some other useful properties of the least-squares fit

1. The sum of the residuals in any regression model that contains an intercept β_0 is always zero that is

$$\sum_{i=1}^n (y_i - \hat{y}_i) = \sum_{i=1}^n e_i = 0$$

2. The sum of the observed values y_i equals the sum of the fitted values $\hat{y}_i$

$$\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$$

3. The least squares regression line always passes through the centroid $[\bar{y}, \bar{x}]$ of the data
4. The sum of the residuals weighted by the corresponding value of the regressor variable always equals zero,

$$\sum_{i=1}^n x_i e_i = 0$$

5. The sum of the residuals weighted by the corresponding fitted value always equals zero, that is,

$$\sum_{i=1}^n y_i e_i = 0$$

Estimation of σ^2

The estimate of σ^2 is obtained from the **residual** or **error sum of squares** defined by

$$SS_{Res} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (7)$$

We can substitute $\hat{y}_i$ in equation 7 by $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ which yields

$$SS_{Res} = \sum_{i=1}^n y_i^2 - n\bar{y}^2 - \hat{\beta}_1 S_{xy}$$

But

$$\sum_{i=1}^n y_i^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2 \equiv SS_T$$

is just the corrected sum of squares of the response observations so

$$SS_{RES} = SS_T - \hat{\beta}_1 S_{xy}$$

The residual sum of squares has $n - 2$ degrees of freedom because two degrees of freedom are associated with the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ involved in obtaining $\hat{y}_i$

An unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_{Res}}{n - 2} = MS_{Res} \quad (8)$$

The quantity MS_{Res} is called the **residual mean square**. The square root of $\hat{\sigma}^2$ is sometimes called the **standard error of regression** and it has the same units as the response variable y

Hypothesis Testing on the Slope and Intercept

Use of t tests

Suppose we wish to test the hypothesis that the slope equals a constant say β_{10}

The appropriate hypothesis are:

$$H_0 : \beta_1 = \beta_{10}, \quad H_1 : \beta_1 \neq \beta_{10}$$

The test statistic is

$$t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{MS_{Res}/S_{xx}}}$$

which follows t_{n-2} distribution if the null hypothesis $H_0 : \beta_1 = \beta_{10}$ is true.

The degrees of freedom associated with t_0 are the number of degrees of freedom associated with MS_{Res} .

This procedure rejects the null hypothesis if

$$|t_0| > t_{\alpha/2, n-2}$$

The denominator of the test statistic t_0 is often called the **estimated standard error** or more simply **standard error** of the slope. That is

$$se(\hat{\beta}_1) = \sqrt{\frac{MS_{Res}}{S_{xx}}}$$

Therefore we often see t_0 written as

$$t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{se(\hat{\beta}_1)}$$

A similar procedure can be used to test the hypothesis about the intercept. To test

$$H_0 : \beta_0 = \beta_{00}, \quad H_1 : \beta_0 \neq \beta_{00}$$

we would use the test statistic

$$t_0 = \frac{\hat{\beta}_0 - \beta_{00}}{\sqrt{MS_{Res}(1/n\bar{x}^2/S_{xx})}} = \frac{\hat{\beta}_0 - \beta_{00}}{se(\hat{\beta}_0)}$$

Example 2 For the data in Example 1 test the hypothesis that

$$H_0 : \beta_1 = 0 \text{ vs } H_1 : \beta_1 \neq 0$$

Solution

We need to compute

$$SS_T = \sum_{i=1}^n y_i^2 - n\bar{y}^2 = 43440958 - 42568974 = 871984$$

The residual sum of squares is

$$\begin{aligned} SS_{Res} &= SS_T - \hat{\beta}_1 S_{xy} \\ &= 871984 - (-22.97 \times -11927.44) \\ &= 871984 - 262403.7 = 609580.3 \end{aligned}$$

Therefore the estimate of σ^2 is computed from

$$\hat{\sigma}^2 = \frac{SS_{Res}}{n-2} = \frac{609580.3}{8} = 76197.54$$

The standard error for $\hat{\beta}_1 = \sqrt{\frac{MS_{Res}}{S_{xx}}} = \sqrt{\frac{76197.54}{519.1557}} = 12.11495$

The test statistic is

$$t_0 = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} = \frac{-22.97}{12.11495} = -1.896$$

If we choose $\alpha = 0.05$ then the critical value is $t_{0.025,8} = 2.306$. Thus we would fail to reject the H_0 and conclude that X has no linear relationship with Y

Analysis of Variance

The analysis of variance is based on a partitioning of total variability in the response variable y .

To obtain this variability begin with the identity

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i) \quad (9)$$

Squaring both sides of equation 9 and summing over all n observations produces

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2 + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y})(y_i - \hat{y}_i)$$

Note that the third term on the right-hand side of this expression can be written as

$$\begin{aligned}
 2 \sum_{i=1}^n (\hat{y}_i - \bar{y})(y_i - \hat{y}_i) &= 2 \sum_{i=1}^n \hat{y}_i(y_i - \hat{y}_i) - 2\bar{y} \sum_{i=1}^n (y_i - \hat{y}_i) \\
 &= 2 \sum_{i=1}^n \hat{y}_i e_i - 2\bar{y} \sum_{i=1}^n e_i = 0
 \end{aligned} \tag{10}$$

Therefore

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

We recognize $SS_{Res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ as the residual sum of squares.

Then $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ the regression sum of squares.

Thus,

$$SS_T = SS_R + SS_{Res}$$

Analysis of variance for testing the significance of regression is given as

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Regression	$SS_R = \hat{\beta}_1 S_{xy}$	1	MS_R	$\frac{MS_R}{MS_{Res}}$
Residual	$SS_{Res} = SS_T - \hat{\beta}_1 S_{xy}$	$n - 2$	MS_{Res}	
Total	SS_T	$n - 1$		

To test the hypothesis that $H_0 : \beta_1 = 0$ compute the test statistic F_0 and reject H_0 if

$$F_0 > F_{\alpha, 1, n-2}$$

Interval Estimation in Simple Linear Regression

Confidence Intervals on β_0, β_1

The sampling distribution of both $(\hat{\beta}_1, \beta_1)/se(\hat{\beta}_1)$ and $(\hat{\beta}_0, \beta_0)/se(\hat{\beta}_0)$ is t with $n - 2$ degrees of freedom.

Therefore a $100(1 - \alpha)$ percent confidence interval on the slope is given by:

$$\hat{\beta}_1 - t_{\alpha/2, n-2} se(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2, n-2} se(\hat{\beta}_1)$$

and a $(100(1 - \alpha))$ percent CI on the intercept β_0 is

$$\hat{\beta}_0 - t_{\alpha/2, n-2} se(\hat{\beta}_0) \leq \beta_0 \leq \hat{\beta}_0 + t_{\alpha/2, n-2} se(\hat{\beta}_0)$$

and consequently the $100(1 - \alpha)\%$ CI on σ^2 is

$$\frac{(n-2)MS_{Res}}{\chi_{\frac{\alpha}{2}, n-2}^2} \leq \sigma^2 \leq \frac{(n-2)MS_{Res}}{\chi_{\frac{1-\alpha}{2}, n-2}^2}$$

```
# reading in the data
Y <- c(2158.70, 1678.15, 2316, 2061.30, 2207.50, 1708.30, 1784.70, 1784.70, 2575, 2357.90)

X <- c(15.50, 23.75, 8.00, 17, 5.5, 19, 24, 2.5, 7.5, 11)

# fitting the model

model <- lm(Y~X)
confint(model)
```

Interval Estimation of Mean Response

Let x_0 be the level of the regressor variable for which we wish to estimate the mean response say $E(y|x_0)$.

We assume that x_0 is any value of the regressor variable within the range of the original data on x used to fit the model.

An unbiased estimator point estimator of $E(Y|x_0)$ is found for the fitted model is:

$$E[\hat{Y}|X_0] = \hat{\mu}_{y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

To obtain a $100(1 - \alpha)$ CI on $E(y|x_0)$, the variance of $\hat{\mu}_{y|x_0}$ is

$$Var(\hat{\mu}_{y|x_0}) = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]$$

The $100(1 - \alpha)$ CI on the mean response at the point $x = x_0$ is

$$\hat{\mu}_{y|x_0} - t_{\frac{\alpha}{2}, n-2} \sqrt{MS_{Res} \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]} \leq E[y|x_0] \leq \hat{\mu}_{y|x_0} + t_{\frac{\alpha}{2}, n-2} \sqrt{MS_{Res} \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]}$$

Prediction of new observations

If x_0 is the value of the regressor variable of interest then

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

is the point estimate of the new value of the response variable y_0

The associated variance is:

$$\sigma^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]$$

The $100(1 - \alpha)\%$ prediction interval on a future observation at x_0 is

$$\hat{y}_0 \pm t_{\frac{\alpha}{2}, n-2} \sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

Example in R using the model

14.95	13.08	12.58	15.97	13.05
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```
# reading in the data
Y <- c(2158.70, 1678.15, 2316, 2061.30, 2207.50, 1708.30, 1784.70, 1784.70, 2575, 2357.90)

X <- c(15.50, 23.75, 8.00, 17, 5.5, 19, 24, 2.5, 7.5, 11)

# fitting the model

model <- lm(Y~X)

# new values

X <- c(14.95, 13.08, 12.58, 15.97, 13.05)

data = data.frame(X)

# predicting the values

pred_model <- predict(model, newdata = data)
pred_model
```

Coefficient of determination

The quantity

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

is called the **coefficient of determination**.

This gives the amount of variation in Y that is explained by X .

```
# reading in the data
Y <- c(2158.70, 1678.15, 2316, 2061.30, 2207.50, 1708.30, 1784.70, 1784.70, 2575, 2357.90)

X <- c(15.50, 23.75, 8.00, 17, 5.5, 19, 24, 2.5, 7.5, 11)

# fitting the model

model <- lm(Y~X)

summary(model)
```

The coefficient of R is given by 0.3143 implying that 31.43% of the variation in Y is explained by X .